दक्षिण बिहार केन्द्रीय विश्वविद्यालय

CENTRAL UNIVERSITY OF SOUTH BIHAR

Department of Computer Science

M. Sc. Computer Science Ist Semester, First Class Test Examination

Subject : Operating Systems **Course Code: CSC81DC00104**

Time : 1 Hours **Date : 22/09/2024** **M. M. : 10**

Note : *Attempt All questions.*

1. What is an Operating System? What are the three main purposes of an OS?

[2]

2. Explain Process Control Block (PCB) and its significance when CPU switches from one process to another.

[2]

3. Consider the following set of processes, with the length of the CPU-burst time given in milliseconds:

Process	Arrival Time	Burst Time	Priority
P1	0	5	1
P2	2	4	3
P3	4	2	5
P4	5	3	1
P5	6	5	2
P6	6	1	4
P7	7	2	9
P8	8	1	3

Draw two Gantt charts illustrating the execution of these processes using preemptive **SJF** and **Priority** scheduling (1 is the highest priority). What is the turnaround time and waiting time of each the processes for both the scheduling algorithms?

[3 +3 =06]

CENTRAL UNIVERSITY OF SOUTH BIHAR

Department of Computer Science

M. Sc. Computer Science Ist Semester, Second Class Test Examination (Date : 18/11/2025)

Subject : Operating Systems **Course Code: CSC81DC00104** **Time : 1 Hours** **M. M. : 10**

Note : *Attempt All questions.*

1. Define the term Deadlock. What are the necessary conditions for a deadlock? [2]
2. Explain the difference between internal and external fragmentation. [2]
3. Explain Banker's algorithm for deadlock avoidance. [3]
4. What is compaction? Explain with example. [3]

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Department of Computer Science

End Term Examination Dec 2025 **Session: 2025-27** **Semester: 1st**

Programme: M.Sc. (CS)	Course Title: Operating Systems	Course Code: CSC81DC00104
Time: 3 Hours	M. M.: 70	Course Credit : 4

Enrollment Number (to be filled by Student):

There are three sections (A, B, and C). Answer all questions as per instructions. Points against each question are mentioned in the parenthesis.

Section – A

Answer All the questions in not more than 50 words. Each question carries 1 Points.
(1 x 10 = 10 Points)

1. Name five major activities of an OS.
2. What are batch systems?
3. What is multitasking?
4. What do you mean by Real Time Operating System (RTOS)?
5. What is difference between the main memory and the secondary memory?
6. What is starvation and aging in OS?
7. What is compaction?
8. What is Thrashing?
9. Explain briefly the Demand Paging.
10. Explains the term Seek Time in context of Disk Scheduling.

Section – B

Answer ANY FIVE questions in not more than 300 words. Each question carries 6 Points.
(6 x 5 = 30 Points)

1. What is an Operating System? What are the three main purposes of an OS?
2. Explain different state,of process units with the help of state transition diagram.
3. Explain context switching of process by giving an example.
4. A dynamic partitioning scheme is being used, and the following is the memory configuration at a given point in time:

1	2	3	4	5	6	7	8	9	10
	30MB	40MB		50MB			60MB		20MB

The areas 1,4,6,7 and 9 are allocated blocks; the others areas are free blocks. The next five memory requests are for 16MB, 24 MB, 20MB, 50MB and 32MB. Which of the following placement algorithm makes the most efficient use of memory?

a. First-fit

b. Best-fit

c. Worst -fit

d. Next-fit. Assume the most recently added block is 8 (60MB) of memory.

[2+1+1+2=06]

5. On a disk with 1000 cylinders, number 0 to 999, compute the number of tracks the disk arm must move to satisfy all the request in the disk queue. Assume the last request was at track 345 and the head is moving towards track 0. The queue in FIFO order contains request for the following tracks: 90,55,125,670,870,105,367 and 490. Perform the computation by using SCAN and LOOK algorithm.

[03+03=06]

6. Write a short note on the following
   a) Process Control Block (PCB)
   b) Race condition

[03+03=06]

Section – C

Answer ANY TWO questions in not more than 1000 words. Each question carries 15 Points.

(15 x 2 = 30 Points)

1. Consider following process, with the CPU burst time given in milliseconds

Process	Burst Time	Arrival Time	Queue Number	Priority
P1	4	1.0	2	5
P2	8	2.0	1	----
P3	5	3.0	3	----
P4	1	4.0	2	10
P5	4	4.0	1	----
P6	6	5.0	2	2
P7	2	5.0	1	-----
P8	2	6.0	2	1
P9	6	8.0	3	---
P10	1	10.0	1	----

The system has three queues. First queue is having the highest priority with SJF scheduling, second queue is having the middle priority with Priority Scheduling (1 is highest and 100 is lowest priority) and, third queue is having the lowest priority with Round Robin Scheduling (T. S. = 2). Draw Gantt charts to show execution using Multi Level Queue scheduling (MLQS). Also determine the process turnaround time and waiting time.

2. Consider the following snapshot of a system:

Available			
R1	R2	R3	R4
3	2	1	0

Process	Current Allocation				Maximum Demand			
	R1	R2	R3	R4	R1	R2	R3	R4
P1	0	0	1	2	0	0	1	2
P2	2	1	0	0	2	4	5	0
P3	0	0	3	4	4	3	4	4
P4	2	2	5	4	4	3	5	6
P5	0	3	3	2	0	6	5	2

Answer the question using the Banker's algorithm:

a) Is the system in a safe state?

b) If a request from process P3 arrives for (0, 1, 0, 0), can the request be granted immediately?

[10+05 =15]

3.

(i) Suppose the OS on your computer uses the buddy system for memory management. Initially the system has a 1 megabyte (1024K) block of memory available, which begins at address 0. Show the results of each request/release via successive figures

A : Request 60 K
B: Request 224K
C: Request 42K
D : Request 56 K
E: Request 30K
F: Request 20K
Release A
Release B
G: Request 40K

After memory is allocated to process G, how much internal fragmentation exists in the system?

(ii) Consider the following page-reference string :

1 2 3 4 ,2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6

How many page faults occur for the following replacement algorithms, assuming three frames?

(i) LRU (ii) FIFO

[07 +08 =15]